

Paper Code: IT604A
Examination – April – 2025

B.Tech. IV Sem : Information Technology
Subject name: Information Security

Time : 3Hrs

Max. Marks : 60

Min. Marks : 19

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c & d has internal Choice. Assume missing data, if any.

Word limit be observed as follows:

Part (a)-Objective type/Fill in the blanks/True & False

Part (b) – Three/four sentences-based answer question (Maximum 50 words)

Part (c) – Short Descriptive Ans./Numerical/ algorithm /programs-based Question.(Maximum 100 words)

Part (d) – Long Answer /Shorts -notes /Descriptive Ans. based Question (Maximum 400 words)

Word limit NOT to be followed for diagram, numerical, derivation.

		Marks	BL	CO
Q.1	(a) (i) The three main principles of security are confidentiality, _____ and availability.	02	1	1
	(ii) _____ key cryptography uses two keys: one for encryption and another for decryption.			
	(b) What are passive attacks?	02	1	1
	(c) List and explain the three main security services.	03	2	1
	OR			
	Explain the concept of symmetric key cryptography with an example.	03	2	1
	(d) Explain substitution and transposition techniques with examples.	05	2	1
	OR			
	What are the common types of attacks in cryptography and how can they be prevented?	05	2	1
Q.2	(a) (i) _____ is a widely used symmetric block cipher algorithm that uses a 56-bit key.	02	1	2
	(ii) _____ is a key exchange protocol that enables two users to securely share a secret over an insecure channel.			
	(b) What is the purpose of a public key in asymmetric encryption?	02	2	2
	(c) What is a digital certificate and how is it used in key distribution?	03	3	2
	OR			
	Explain the role of RC4 in symmetric encryption	03	3	2
	(d) Compare and contrast stream ciphers and block ciphers.	05	4	2
	OR			
	Compare RSA, ECC, and Diffie-Hellman in terms of security and performance.	05	4	2
Q.3	(a) (i) The _____ algorithm is a widely used cryptographic hash function that produces a 160-bit hash value.	02	1	3
	(ii) In Kerberos, the _____ distributes session keys and tickets for authentication.			
	(b) Define digital signature	02	1	3
	(c) Explain the purpose and working of Public Key Infrastructure (PKI).	03	2	3
	OR			
	Describe the concept and one limitation of the knapsack algorithm.	03	2	3

(d) Compare and contrast HMAC and CMAC. Provide use-case examples. 05

4	3
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OR

Describe how biometric authentication works and highlight its advantages and challenges. 05

4	3
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- Q.4 (a) (i) PGP ensures message integrity by creating a _____ of the original message. 02

1	4
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- (ii) _____ Protocol is commonly used for automated key management in IPsec.
- (b) What is the role of a Certificate Authority (CA) in S/MIME? 02

1	4
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- (c) Compare the Authentication Header (AH) and Encapsulating Security Payload (ESP) in IPsec. 03

4	4
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OR

How does IPsec ensure the integrity and confidentiality of data sent over the internet? 03

4	4
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- (d) Explain the working of Pretty Good Privacy (PGP) for email security, including how encryption, signing, and key management are handled. 05

3	4
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OR

Compare and contrast PGP and S/MIME in terms of their features, implementation, and use cases for email security. 05

3	4
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- Q.5 (a) (i) **SSL** stands for _____ and is used to secure communication over the internet. 02

1	5
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- (ii) In virtual elections, _____ is crucial to maintain voter privacy, election integrity, and to prevent tampering.
- (b) What is the primary function of **Secure Socket Layer (SSL)** in web security? 02

2	5
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- (c) How does a **firewall** function to protect a network from unauthorized access? 03

3	5
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OR

Describe the **process of Secure Inter-branch Payment Transactions** using encryption. 03

3	5
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- (d) Explain the key differences between the **types of firewalls** and provide examples of their use in securing networks. 05

4	5
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OR

Analyze the case study of **Secure Inter-branch Payment Transactions** and explain how cryptographic techniques, such as encryption and digital signatures, ensure secure transactions. 05

4	5
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Bloom's Taxonomy Levels (BL): 1-Remembering, 2- Understanding, 3- Applying, 4- Analysing, 5-Evaluating, 6-Creating
